

Competitive Exam Practice 5: Addressing Modes, CPU Organization, and Bus Structures

GATE (Computer Science) Pattern

PCC CS-401: Computer Organization and Architecture

August 7, 2026

Each question carries 1–2 marks, consistent with GATE CS's Computer Organization & Architecture weighting. MCQ = single correct option; NAT = Numerical Answer Type (enter a numeric value, no options given).

- Q1** [*Original, GATE-pattern*] [**MCQ, 1 mark**] Which addressing mode is most suitable for accessing successive elements of an array inside a loop whose index variable changes on every iteration?
- (A) Immediate
 - (B) Direct
 - (C) Displacement (indexed)
 - (D) Register
- Q2** [*Original, GATE-pattern*] [**MCQ, 2 marks**] An instruction uses indirect addressing with address field A . If $M[A] = B$ and $M[B] = 42$, the effective address of the operand is:
- (A) A
 - (B) B
 - (C) 42
 - (D) $M[42]$
- Q3** [*Original, GATE-pattern*] [**NAT, 1 mark**] An instruction `LOAD R1, 25(R4)` uses displacement addressing, with $[R4] = 175$. The effective address is _____.
- Q4** [*Original, GATE-pattern*] [**MCQ, 1 mark**] On a single-bus CPU datapath, a two-operand ALU instruction (such as `ADD R1, R2, R3`) that must stage both operands sequentially through the bus requires a minimum of how many control steps?
- (A) 1
 - (B) 2
 - (C) 3
 - (D) 4
- Q5** [*Original, GATE-pattern*] [**MCQ, 2 marks**] In a three-bus CPU organization where Bus A and Bus B feed the ALU's two inputs and Bus C carries the ALU's output, the primary architectural advantage over a single-bus organization is:

- (A) Reduced control unit complexity
- (B) Fewer registers required in the register file
- (C) Two operands can be read and the result written back, all within the same clock cycle
- (D) Lower total hardware cost

Q6 [*Original, GATE-pattern*] [**NAT, 2 marks**] Indirect addressing (via a pointer stored in memory, not in a register) requires _____ memory access(es) in total to obtain the operand.